TALLER 2 – CONTENERIZACIÓN DE APP CON DOCKER

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GITHUB: https://github.com/fernanda419/taller-docker.git

Barranquilla- Atlántico

2026

Sena

Calentamiento: comprobar que Docker funciona

evidencia 1

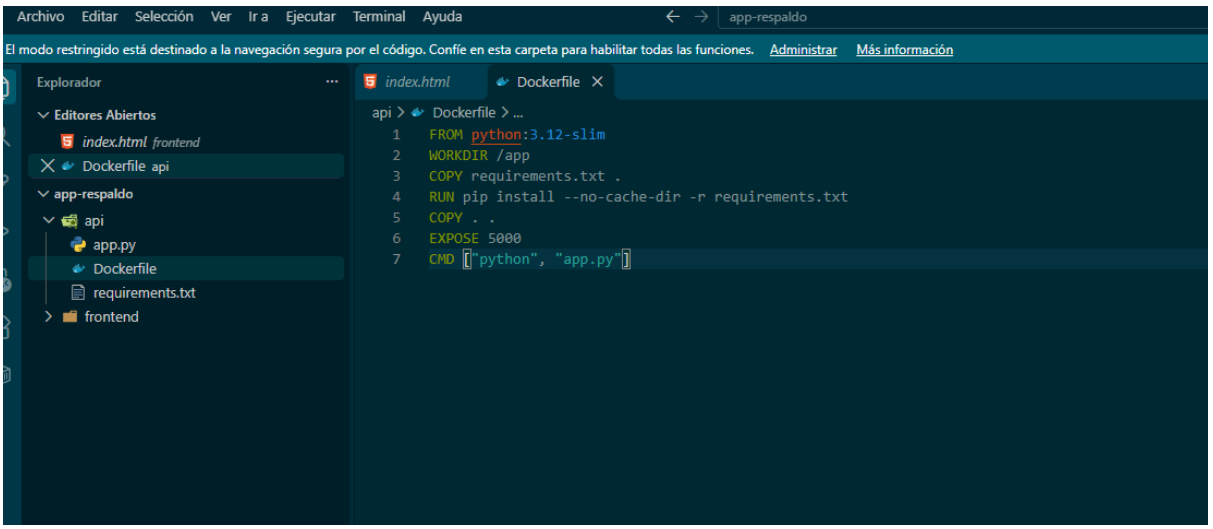
```
PS C:\WINDOWS\system32> docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS                                     NAMES
68b0bd6622e9   nginx     "/docker-entrypoint.…"   2 minutes ago   Up 2 minutes   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   prueba-web
PS C:\WINDOWS\system32> docker logs prueba-web
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/10/14 18:28:52 [notice] 1#1: using the "epoll" event method
2026/10/14 18:28:52 [notice] 1#1: nginx/1.31.6
2026/10/14 18:28:52 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/10/14 18:28:52 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/10/14 18:28:52 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/10/14 18:28:52 [notice] 1#1: start worker processes
2026/10/14 18:28:52 [notice] 1#1: start worker process 29
2026/10/14 18:28:52 [notice] 1#1: start worker process 30
2026/10/14 18:28:52 [notice] 1#1: start worker process 31
2026/10/14 18:28:52 [notice] 1#1: start worker process 32
2026/10/14 18:28:52 [notice] 1#1: start worker process 33
2026/10/14 18:28:52 [notice] 1#1: start worker process 34
2026/10/14 18:28:52 [notice] 1#1: start worker process 35
2026/10/14 18:28:52 [notice] 1#1: start worker process 36
172.17.0.1 - - [14/Oct/2026:18:28:52 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (
2026/10/14 18:28:53 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0
//localhost:8080/"
172.17.0.1 - - [14/Oct/2026:18:28:53 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0;
PS C:\WINDOWS\system32>
```

```
PS C:\WINDOWS\system32> docker rm -f prueba-web
prueba-web
PS C:\WINDOWS\system32>
```

Preparar el proyecto y escribir el Dockerfile de la API

```
Windows PowerShell
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PS C:\Users\FabricaT.I.C> Expand-Archive -Path "$HOME\Downloads\app-respaldo.zip" -DestinationPath C:\docker-
PS C:\Users\FabricaT.I.C> Expand-Archive -Path "$HOME\Downloads\app-respaldo.zip" -DestinationPath C:\dockeradso\taller
PS C:\Users\FabricaT.I.C> cd C:\dockeradso\taller\app-respaldo
PS C:\dockeradso\taller\app-respaldo> code .
PS C:\dockeradso\taller\app-respaldo>
```




```
PS C:\dockeradso\taller\app-respaldo> cd api
PS C:\dockeradso\taller\app-respaldo\api> docker build -t tareas-api:1.0 .
[+] Building 14.7s (10/10) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 200B
 => [internal] load metadata for docker.io/library/python:3.12-slim
 => [internal] load .dockerignore
 => => transferring context: 74B
 => [1/5] FROM docker.io/library/python:3.12-slim@sha256:05cda9777409a9c3ffddd94a4c476b79f0769a0b4857f0c7ed9226b6800b0d6f
 => => resolve docker.io/library/python:3.12-slim@sha256:05cda9777409a9c3ffddd94a4c476b79f0769a0b4857f0c7ed9226b6800b0d6f
 => [internal] load build context
 => => transferring context: 2.42kB
 => CACHED [2/5] WORKDIR /app
 => CACHED [3/5] COPY requirements.txt .
 => [4/5] RUN pip install --no-cache-dir -r requirements.txt
 => [5/5] COPY . .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:9371bf81210814b4c40f0e3d6835462e433b0dc66c828248fdc029c46628ddd0
 => => exporting config sha256:26da386ba85b76f782657169b6156b8ccb9104842e56f8fb78c0eca65a8ced4e
 => => exporting attestation manifest sha256:7b3887b6f82304b71d98d4c95d4acf5129cc2c8bb560cceeef5f8962e8dc84c4
 => => exporting manifest list sha256:be5700f9d9e4e00d946387610dea6a9d651652ec8c040f2d476ca7b0747b0ac6
 => => naming to docker.io/library/tareas-api:1.0
 => => unpacking to docker.io/library/tareas-api:1.0

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/0jmydvyuh18m5r1u2w8q3xd71
PS C:\dockeradso\taller\app-respaldo\api>
```

evidencia 2

```
PS C:\dockeradso\taller\app-respaldo\api> docker run -d -p 5000:5000 --name api-prueba `
>> -e DB_HOST=bd -e DB_USER=x -e DB_PASSWORD=x -e DB_NAME=x `
>> tareas-api:1.0
61a8181edad5f3cf9b805ef16e2c036c8ae1de54670349d45c3be4288bd53099
PS C:\dockeradso\taller\app-respaldo\api> docker logs api-prueba
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://172.17.0.2:5000
Press CTRL+C to quit
PS C:\dockeradso\taller\app-respaldo\api>
```

Hacer un cambio y sacar la versión 1.1

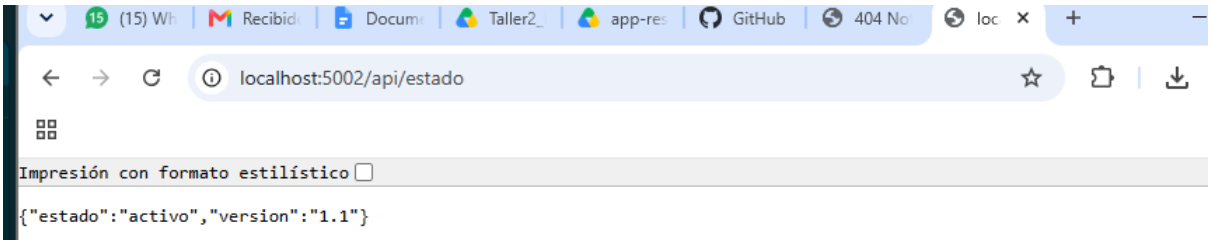
evidencia 3

```
PS C:\dockeradso\taller\app-respaldo\api> docker images tareas-api
                                                                    Info →
IMAGE              ID             DISK USAGE   CONTENT SIZE   EXTRA
tareas-api:1.0     be5700f9d9e4         217MB         53.1MB
tareas-api:1.1     9beaa09402ae         217MB         53.1MB
PS C:\dockeradso\taller\app-respaldo\api> docker run -d -p 5001:5000 --name api-v10 `
>> -e DB_HOST=bd -e DB_USER=x -e DB_PASSWORD=x -e DB_NAME=x tareas-api:1.0
38756c0e4490231f04db6bcf7bf91b1750d8206218a5fa97aa37a0c8f1fffda3
PS C:\dockeradso\taller\app-respaldo\api> docker run -d -p 5002:5000 --name api-v11 `
>> -e DB_HOST=bd -e DB_USER=x -e DB_PASSWORD=x -e DB_NAME=x tareas-api:1.1
4d9713e9239b80d7467ae0a54c2f63382a5414ffc0e3eac2629dae2949ea2a27
PS C:\dockeradso\taller\app-respaldo\api>
```



Not Found

The requested URL was not found on the server. If you entered the URL manually please check your spelling and try again.



Comprobar que la base de datos no pierde la información

evidencia 4

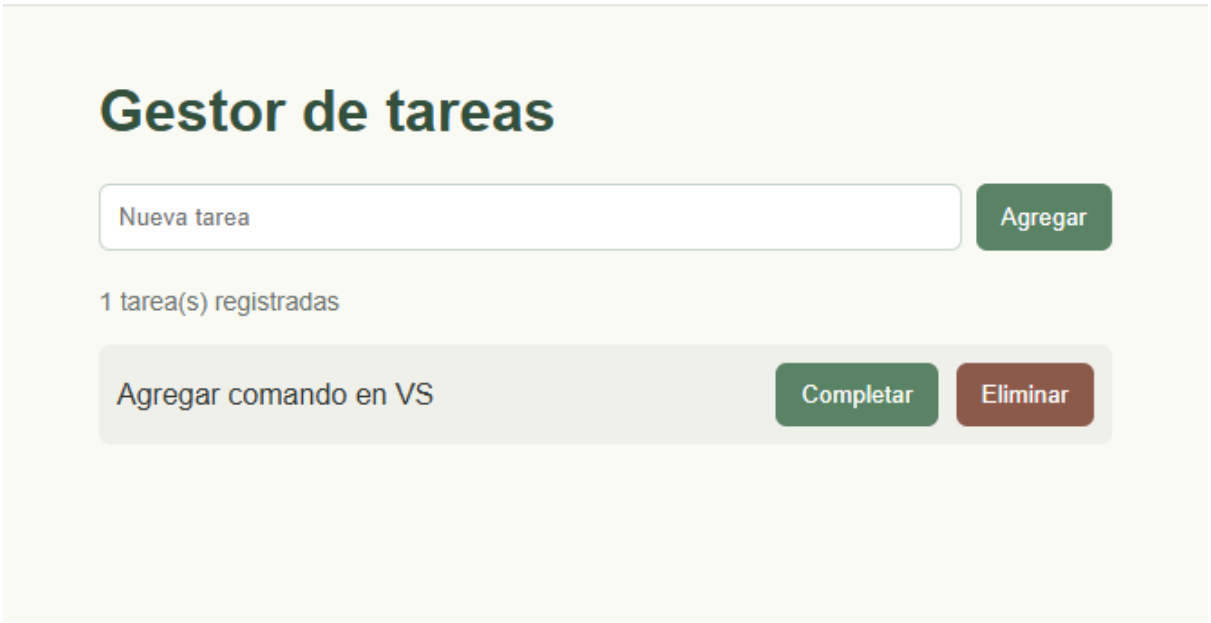
```
PS C:\dockeradso\taller\app-respaldo\api> docker rm -f bd-taller
bd-taller
PS C:\dockeradso\taller\app-respaldo\api> docker run -d --name bd-taller `
>> -e MYSQL_ROOT_PASSWORD=ClaveTaller2026 `
>> -v datos-taller:/var/lib/mysql `
>> mysql:8.4
62ca71c3df7fe153535623b9acfbdb756f9bc7c6837e63d214014982cd1e3a46
PS C:\dockeradso\taller\app-respaldo\api> docker exec -it bd-taller mysql -uroot -pClaveTaller2026 taller `
>> -e "SELECT * FROM prueba;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+----+----------------------+
| id | dato                 |
+----+----------------------+
|  1 | Registro persistente |
+----+----------------------+
PS C:\dockeradso\taller\app-respaldo\api>
```

Unir página web, API y base de datos con Docker Compose

evidencia 5


Gestor de tareas
Nueva tarea
Agregar
2 tarea(s) registradas
Agregar comando en VS
Completar
Eliminar
Terminar proyecto de docker
Completar
Eliminar


localhost:8080

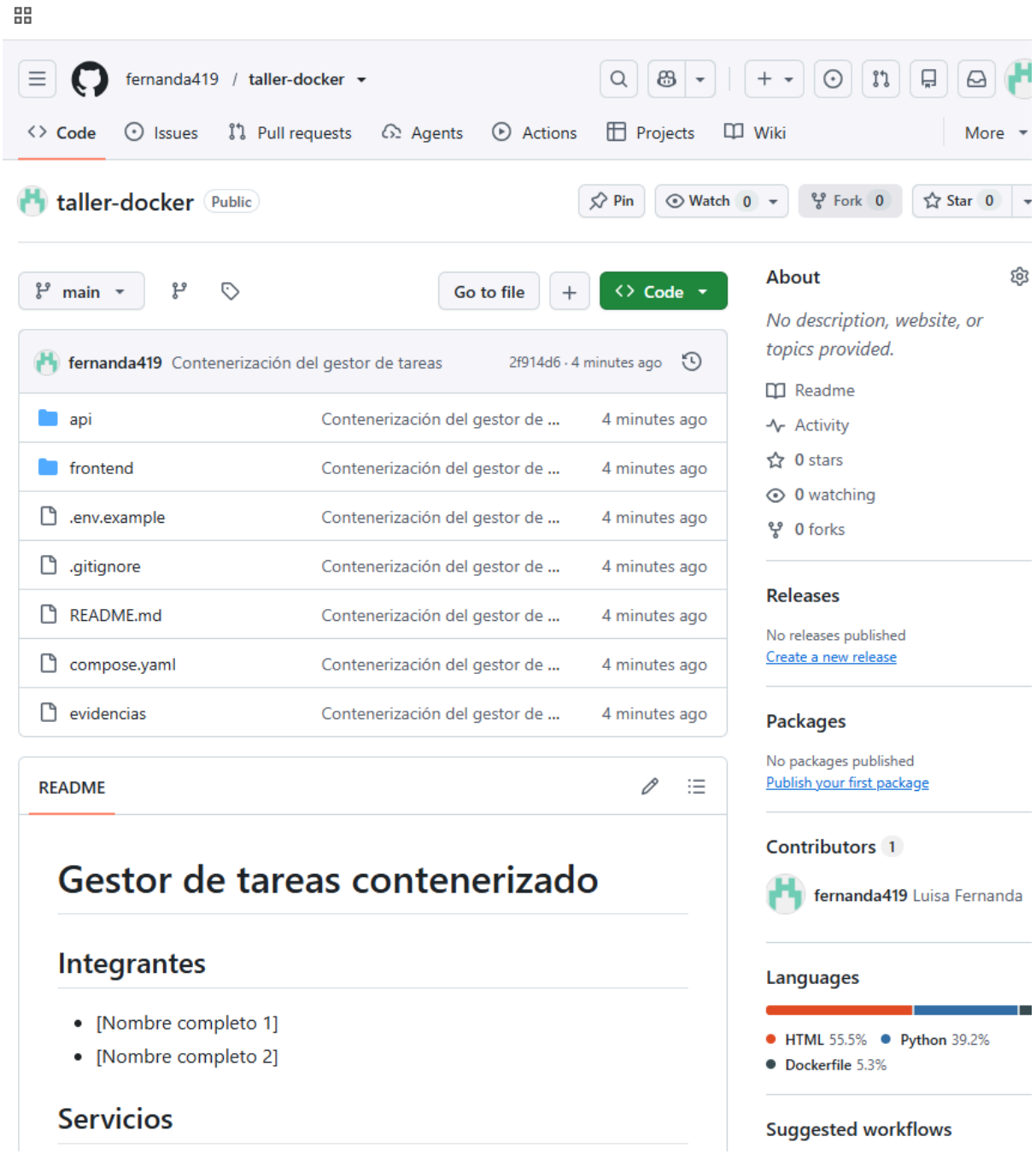

Gestor de tareas
Nueva tarea
Agregar
1 tarea(s) registradas
Agregar comando en VS
Completar
Eliminar

```
PS C:\dockeradso\taller\app-respaldo> docker compose down
[+] Running 4/4
 ✓ Container app-respaldo-frontend-1  Removed
 ✓ Container app-respaldo-api-1       Removed
 ✓ Container app-respaldo-bd-1        Removed
 ✓ Network app-respaldo_default       Removed
PS C:\dockeradso\taller\app-respaldo> docker compose up -d
[+] Running 3/4
 ✓ Network app-respaldo_default       Created
 ✓ Container app-respaldo-bd-1        Started
 - Container app-respaldo-api-1       Starting
 ✓ Container app-respaldo-frontend-1  Created
```

```
PS C:\dockeradso\taller\app-respaldo> docker compose ps
NAME                      IMAGE                  COMMAND                  SERVICE    CREATED
STATUS              PORTS
app-respaldo-api-1        app-respaldo-api       "python app.py"          api        About a minute ago
Up About a minute   5000/tcp
app-respaldo-bd-1         mysql:8.4              "docker-entrypoint.s…"   bd         About a minute ago
Up About a minute   3306/tcp, 33060/tcp
app-respaldo-frontend-1   app-respaldo-frontend  "/docker-entrypoint.…"   frontend   About a minute ago
Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
PS C:\dockeradso\taller\app-respaldo>
```



Documentar, publicar en GitHub y probar con otro equipo



https://github.com/fernanda419/taller-docker.git